CSE STUDY HUB

Communication Skills for Engineers II — Aptitude

UNIT 2 · Time-Based Reasoning

What this unit is about

This unit builds understanding of **Time-Based Reasoning**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Calendar basics: days, weeks, months and leap-year rule

2. Ordinary year vs leap year; odd days

3. Finding day of the week using odd-day counting

4. Clock dial, hour hand and minute hand speeds

5. Angle between hands: |30H – 5.5M| degrees

6. Coincidence, right angles and opposite positions

7. Faulty clocks: gain/loss and correct-time comparison

8. Interchange of hands and interpreting incorrect readings

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 2 NOTES · PART 1

Calendar basics: days, weeks, months and leap-year rule

An ordinary year has 365 days (1 odd day); a leap year has 366 days (2 odd days). A leap year is divisible by 4, except century years must also be divisible by 400.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Ordinary year vs leap year; odd days

Compare the alternatives in Ordinary year vs leap year; odd days using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Finding day of the week using odd-day counting

Finding day of the week using odd-day counting is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Time-Based Reasoning, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 2

Clock dial, hour hand and minute hand speeds

The minute hand moves 6° per minute and the hour hand 0.5° per minute. At H:M, the smaller angle is based on |30H–5.5M|, adjusted to at most 180°.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Angle between hands: |30H − 5.5M| degrees

For Angle between hands: |30H – 5.5M| degrees, identify the given values and the quantity to find before selecting a formula. Keep units consistent, show substitutions, calculate carefully and perform a reasonableness check. Write the formula and one solved example in your revision copy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Coincidence, right angles and opposite positions

The OSI model has Physical, Data Link, Network, Transport, Session, Presentation and Application layers. Each layer provides services to the one above and uses the one below; encapsulation adds headers as data moves down.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Time-Based Reasoning, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 3

Faulty clocks: gain/loss and correct-time comparison

The minute hand moves 6° per minute and the hour hand 0.5° per minute. At H:M, the smaller angle is based on |30H–5.5M|, adjusted to at most 180°.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Interchange of hands and interpreting incorrect readings

Interchange of hands and interpreting incorrect readings is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Time-Based Reasoning, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Calendar basics: days, weeks, months and leap-year rule** — know the definition, core working and one example.

• **Ordinary year vs leap year; odd days** — know the definition, core working and one example.

• **Finding day of the week using odd-day counting** — know the definition, core working and one example.

• **Clock dial, hour hand and minute hand speeds** — know the definition, core working and one example.

• **Angle between hands: |30H − 5.5M| degrees** — know the definition, core working and one example.

• **Coincidence, right angles and opposite positions** — know the definition, core working and one example.

• **Faulty clocks: gain/loss and correct-time comparison** — know the definition, core working and one example.

• **Interchange of hands and interpreting incorrect readings** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Time-Based Reasoning. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Calendar basics: days, weeks, months and leap-year rule and explain its importance.

Q2. Explain the working/steps involved in Ordinary year vs leap year; odd days.

Q3. Differentiate between two related concepts from Time-Based Reasoning.

Q4. Give one practical application of Interchange of hands and interpreting incorrect readings and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Time-Based Reasoning.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.